

# STATEMENT OF PURPOSE

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Ph.D. in Statistics

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**My Academic Goal.** Through my experiences in various industrial projects and applied studies, I have confronted a stark reality. Data is never as clean as textbook theory suggests. It is always the product of complex, imperfect mechanisms, with information fragmented throughout its structure. To me, this is not a limitation. It is the central challenge for statistics to solve. My academic goal is to develop theories that perform in this complexity, rather than remaining as abstract proofs. As a first step toward this goal, my focus turned to graphical models and optimal transport, essential for handling intricate data structures.

**Graphical Models Under Data Contamination.** My interest in graphical models emerged from repeated experiences in applied projects where key variable relationships were obscured by various forms of data imperfection, such as measurement error, miscoding, or missingness. These settings made clear that real-world data rarely reveal their underlying structure cleanly, yet understanding such structure is often essential. Motivated by this gap, I pursued two complementary studies [1, 2] on causal discovery under contamination.

In [2], I explored this idea in the Gaussian setting, showing that contamination systematically inflates the apparent connectivity of a graph. This insight led to a principled strategy deeply aligned with Occam’s razor: among all candidate structures obtained after correcting for variability, the simplest (i.e., sparsest) one reveals the true relationships. This approach is powerful because it works even when the contamination parameters are unknown. However, I was driven to move beyond theories confined to the Gaussian regime, seeking to generalize these principles.

I tackled this limitation in [1] by developing a distribution-free approach. By studying how the geometry of moments encodes structure, I shifted the focus from traditional conditional independence to linear orthogonality. This allowed me to formulate a distribution-free condition connecting moment-based orthogonality to graph separation, which in turn enabled a consistent recovery algorithm for a much broader class of models. The trade-off was that this generality required the contamination process to be known, presenting a complementary challenge to my first study.

Through these studies, I became fascinated by how structural information can persist—sometimes even sharpen—under data contamination. This perspective now guides my broader interest in developing robust statistical methods that remain reliable even when observations are imperfect.

**Optimal Transport Between Metric Spaces.** While my previous work treated graphs as a way to reveal relationships between variables, graphs are also a powerful representation for the data itself, especially in non-Euclidean domains. A core scientific challenge is to match these seemingly different spaces, such as individual brain connectomes or molecular structures, to uncover their shared characteristics. The Gromov–Wasserstein (GW) and fused GW frameworks provide elegant tools for this matching task. However, their reliance on highly non-convex objectives leads to unstable solutions, complicating theoretical analysis and hindering reliable inference.

My current research draws inspiration from convex relaxation techniques in graph matching, generalizing this approach from the specific graph matching problem to the broader challenge of comparing metric spaces. This generalization required a novel extension of convex relaxation principles

to the operator level, which I then used to design a new fused optimal transport formulation that is both convex and geometry-preserving. This fused loss is effective since it simultaneously considers feature similarity and geometric alignment. More than just a computational benefit, the convexity of this approach guarantees a global optimum, providing a crucial stability that allowed me to provide finite-sample consistency guarantees for a geometry-aware fused optimal transport method. This research provides a compelling algorithm for matching complex non-Euclidean data, representing a key step toward my goal of building reliable theories for intricate data.

**My Blueprint: Distilling Information from Massive and Intricate Data.** While modern AI often relies on massive models and large-scale computation, I believe that the essential information in data is fundamentally small and structured. My goal is to design statistical algorithms that identify and utilize this structure, enabling both statistical and computational efficiency.

My research interests are now moving toward addressing the critical need for high-dimensional selection inference under contamination. As complex data sources become ubiquitous, we need principled algorithms that detect key variables while remaining robust to noise and adversarial perturbations. I plan to develop such robust selection methods with concrete guarantees, a direction that extends naturally from my work on identifying structures under imperfect observations.

This pursuit of essential information naturally scales up to another area I am passionate about: domain adaptation and transfer learning. Many modern statistical tasks require transferring knowledge across domains that are heterogeneous yet share a common underlying structure. My work in geometry-preserving optimal transport provides a powerful foundation here, as I aim to use geometric alignment as a principled mechanism to extract the invariant, common information that allows for generalization. This connects optimal transport with distributional robustness, offering a new lens for learning from diverse sources.

Ultimately, these interests converge on the central goal of building a unified and practical framework for statistical learning. My research experience has prepared me to tackle this objective, which aims to yield both rigorous theories and practical tools for complex systems.

**Why UC Davis.** UC Davis offers the ideal environment to advance my goal of developing principled statistical methods for complex and imperfect data. I am particularly drawn to the work of **Prof. OOO**, whose research on optimization, geometric statistics, and high-dimensional inference directly supports my academic goal of developing statistical methods for complex data. Likewise, **Prof. OOO**'s contributions to graphical models and variable screening provide a natural foundation for extending my work on robust graphical models toward contamination-resilient selective inference. In parallel, **Prof. OOO**'s expertise in optimization and entropic optimal transport speaks directly to my interest in geometry-aware optimal transport. Her perspective at designing efficient optimization technique would substantially enrich my training at the intersection of statistics and computation.

My post-graduation goal is to pursue an academic career devoted to building a unified and practical framework for statistical learning. To advance this vision, I hope to train in an environment that integrates robustness, geometric reasoning, and high-dimensional statistical theory into a coherent intellectual foundation, where I believe UC Davis embodies exactly this combination.

## My Publications

- [1] **Junhyoung Chung**<sup>\*</sup>, Youngmin Ahn, Donguk Shin, and Gunwoong Park. “Learning distribution-free anchored linear structural equation models in the presence of measurement error”. In: *Journal of the Korean Statistical Society* (2024), pp. 1–25.
- [2] Joonho Shin<sup>†</sup>, **Junhyoung Chung**<sup>†</sup>, Seyong Hwang<sup>†</sup>, and Gunwoong Park<sup>†</sup>. “Discovering Causal Structures in Corrupted Data: Frugality in Anchored Gaussian DAG Models”. In: *Computational Statistics and Data Analysis* (2025), p. 108267.

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